

ON THE ADJUNCTION BETWEEN GEOMETRIC REALISATION AND SINGULAR SET

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ABSTRACT. These are expository notes on simplicial sets. It contains a proof that geometric realisation of a simplicial set and singular set are adjoint functors; and that the full and diagonal realisation of a bisimplicial set are homeomorphic. No originality is claimed there.

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1. INTRODUCTION

The creation of these notes starts from a problem given by Professor Daniel Ramras, that the full realisation and the diagonal realisation of a bisimplicial set are homeomorphic (Corollary 5.0.2) by showing that they have the bisingular set as adjoint functor (Theorem 5.0.1). But then, we realised that it was also a good occasion to keep a record on what we have

learnt about simplicial sets, as they are inevitable when one is learning algebraic K -theory [Qui73; Wal85].

The primary purpose of these notes is to provide a detailed proof that geometric realisation and singular set are adjoint functors, and its analogues for bisimplicial sets¹ (Theorem 5.0.1), hence two topological spaces created from a bisimplicial set are homeomorphic. Secondly, we hope these notes can serve as a quick reference to remind readers the proofs of some fundamental results of simplicial sets. The readership, however, is not pleasant at all—it requires reader to understand some algebraic topology (say, singular homology at a minimum) and category theory (Yoneda’s Lemma, colimit of functors, adjoint functor, etc. Mac Lane’s famous book [ML98] is definitely more than enough.). Greg Friedman’s notes [Fri08] is an excellent choice for beginners. We try to follow his spirit when preparing these notes.

There are many highlights when one is learning simplicial set—the title of these notes (Theorem 2.8.1), Milnor’s theorem on the geometric realisation of product of simplicial sets ([Mil57, Theorem 2 on page 359]), just to name a few. We try to prove all results used in this paper. If not, we provide a direction to the proof in the references to satisfy readers’ curiosities.

1.1. Convention. Throughout this paper the term “functor” is always assumed to mean “covariant functor”.

2. SIMPLICIAL SETS

2.1. The category Δ . For each non-negative integer n , we denote $[n]$ to be the set

$$[n] := \{0, 1, 2, \dots, n\} \subseteq \mathbb{N} \cup \{0\}.$$

Then, we define the category Δ whose objects are $[n]$, and $\text{Mor}_\Delta([n], [m])$ consists of order-preserving maps under the standard ordering of $\mathbb{Z}$.

We introduce two important morphisms in Δ . For each $0 \leq i \leq n$, define the **coface map** $d^i : [n-1] \rightarrow [n]$ and the **codegeneracy map** $s^i : [n+1] \rightarrow [n]$ by

$$d^i(j) := \begin{cases} j & \text{if } j < i; \\ j+1 & \text{if } j \geq i, \end{cases} \quad s^i(j) := \begin{cases} j & \text{if } j \leq i; \\ j-1 & \text{if } j > i. \end{cases}$$

It is a straight forward verification that the cofaces and codegenerates satisfy the following **cosimplicial identities**:

$$\begin{cases} d^j d^i = d^i d^{j-1} & \text{if } i < j, \\ s^j d^i = d^i s^{j-1} & \text{if } i < j, \\ s^j d^j = \text{id} = s^j d^{j+1}, \\ s^j d^i = d^{i-1} s^j & \text{if } i > j+1, \\ s^j s^i = s^i s^{j+1} & \text{if } i \leq j. \end{cases} \quad (2.1)$$

Clearly, cofaces (resp. codegenerates) are injective (resp. surjective). Our next statement is that the cofaces and codegenerates generate all morphisms in Δ .

¹so that professors will have to find harder problems for their research students.

Lemma 2.1.1 ([Wei95, Lemma 8.1.2 on page 255])

Every morphism $\alpha : [n] \rightarrow [m]$ in Δ has a unique epi-monic factorisation $\alpha = DS$, where D (resp. S) is uniquely a composition of cofaces (resp. codegenerates).

Observe that there is a functor

$$\text{Simplex} : \Delta \rightarrow \mathbf{Top} \quad (2.2)$$

defined by sending $[n]$ to the geometric n -simplex

$$|\Delta^n| := \left\{ (t_0, \dots, t_n) \in \mathbb{R}^{n+1} \mid \sum_{i=0}^n t_i = 1, \text{ and } t_i \geq 0 \right\},$$

and

$$\begin{aligned} (d^i : [n] \rightarrow [n+1]) &\mapsto (D_i : |\Delta^n| \rightarrow |\Delta^{n+1}|), \\ (s^i : [n] \rightarrow [n-1]) &\mapsto (S_i : |\Delta^n| \rightarrow |\Delta^{n-1}|), \end{aligned}$$

which are explicitly given by the formulae

$$\begin{aligned} D_i(t_0, \dots, t_n) &:= (t_0, \dots, t_{i-1}, 0, t_i, \dots, t_n), \\ S_i(t_0, \dots, t_n) &:= (t_0, \dots, t_i + t_{i+1}, t_{i+2}, \dots, t_n) \end{aligned} \quad (2.3)$$

2.2. Simplicial Sets. Let $\mathbf{Set}$ be the category of sets. A simplicial set is a functor $X : \Delta^{op} \rightarrow \mathbf{Set}$. We will abbreviate $X_n := X([n])$, and call it the set of n -simplices of X

For each coface and codegeneracy we have the corresponding face map $d_i := X(d^i) : X_n \rightarrow X_{n-1}$ and degeneracy map $s_i := X(s^i) : X_n \rightarrow X_{n+1}$, satisfying the **simplicial identities**

$$\begin{cases} d_i d_j = d_{j-1} d_i & \text{if } i < j, \\ d_i s_j = s_{j-1} d_i & \text{if } i < j, \\ d_j s_j = \text{id} = d_{j+1} s_j, \\ d_i s_j = s_j d_{i-1} & \text{if } i > j+1, \\ s_i s_j = s_{j+1} s_i & \text{if } i \leq j. \end{cases} \quad (2.4)$$

We will sometimes use d_i^X to denote the i -th face map for the simplicial set X if multiple simplicial sets are involved, and similarly for degeneracy maps.

We say an n -simplex $x_n \in X_n$ is **non-degenerate** if $x_n \neq s_i x_{n-1}$ for any i and any $x_{n-1} \in X_{n-1}$.

2.3. Examples of Simplicial Sets.

2.3.1 We define $\Delta^n : \Delta^{op} \rightarrow \mathbf{Set}$ to be the Hom-set

$$\Delta_k^n := \text{Hom}_\Delta([k], [n]),$$

with face (resp. degeneracy) maps defined to be composition with coface (resp. codegeneracy) maps:

$$\begin{aligned} d_j^{\Delta^n}(\varphi : [k] \rightarrow [n]) &:= \varphi \circ d^j \\ s_j^{\Delta^n}(\varphi : [k] \rightarrow [n]) &:= \varphi \circ s^j \end{aligned}$$

2.3.2 (**Singular Set**) Given a topological space Y , the **singular set** $\mathcal{S}(Y)$ associated to Y is a simplicial set with

$$\begin{aligned} \mathcal{S}(Y)_n &:= \{\text{singular } n\text{-simplex of } Y\} \\ &= \{\sigma : |\Delta^n| \rightarrow Y \mid \sigma \text{ is continuous}\} \\ &=: \text{Hom}_{\mathbf{Top}}(|\Delta^n|, Y) \end{aligned}$$

For any singular n -simplex σ , the face and degenerate maps for are defined as

$$\begin{aligned} d_i^{\mathcal{S}(Y)}\sigma &:= \sigma \circ D_i \\ s_i^{\mathcal{S}(Y)}\sigma &:= \sigma \circ S_i \end{aligned}$$

The maps D_i and S_i are defined as in Equation (2.3).

2.3.3 (**Product of Simplicial Sets**) Given two simplicial sets X, Y , their **product** $X \times Y$ is a simplicial set defined by

$$\begin{aligned} (X \times Y)_n &:= X_n \times Y_n, \\ d_i^{X \times Y}(x_n, y_n) &:= (d_i^X x_n, d_i^Y y_n), \\ s_i^{X \times Y}(x_n, y_n) &:= (s_i^X x_n, s_i^Y y_n), \end{aligned}$$

for $x_n \in X_n$ and $y_n \in Y_n$.

2.3.4 Let $\mathcal{C}$ be a small category. Its **nerve** $\mathcal{N}(\mathcal{C})$ is the simplicial set defined by

$$\begin{aligned} \mathcal{N}_0(\mathcal{C}) &:= \text{Obj}(\mathcal{C}), \\ \mathcal{N}_n(\mathcal{C}) &:= \left\{ C_0 \xrightarrow{f_0} C_1 \xrightarrow{f_1} \dots \xrightarrow{f_{n-1}} C_n \mid \begin{array}{l} f_i, f_{i-1} \text{ are composable} \\ \text{for } i = 1, \dots, n. \end{array} \right\} \quad \text{for } n \geq 1. \end{aligned}$$

The i -th face map $d_i^{\mathcal{N}(\mathcal{C})} : \mathcal{N}_n(\mathcal{C}) \rightarrow \mathcal{N}_{n-1}(\mathcal{C})$ is given by

$$d_i^{\mathcal{N}(\mathcal{C})} \left(C_0 \xrightarrow{f_0} \dots \xrightarrow{f_{n-1}} C_n \right) := \left(C_0 \xrightarrow{f_0} \dots \xrightarrow{f_{i-2}} C_{i-1} \xrightarrow{f_i \circ f_{i-1}} C_{i+1} \xrightarrow{f_{i+2}} \dots \xrightarrow{f_{n-1}} C_n \right);$$

and the i -th degeneracy map is given by

$$s_i^{\mathcal{N}(\mathcal{C})} \left(C_0 \xrightarrow{f_0} \dots \xrightarrow{f_{n-1}} C_n \right) := \left(C_0 \xrightarrow{f_0} \dots \xrightarrow{f_{i-1}} C_i \xrightarrow{\text{id}_{C_i}} C_i \xrightarrow{f_i} \dots \xrightarrow{f_{n-1}} C_n \right).$$

The notion of the nerve of a category plays an important role in Quillen's Q -construction of higher algebraic K -theory [Qui73, §2].

2.4. Simplicial Maps and the Category of Simplicial Sets. A **simplicial map** $f : X \rightarrow Y$ between simplicial sets is a natural transformation of these functors. Thus simplicial sets form a category, which we will denote it as **sSet**. One can also check that the product of simplicial sets defined in Example 2.3.3 is the categorical product in **sSet** by verifying the universal property directly.

The following lemma is easily checked:

Lemma 2.4.1

Let X, Y be two simplicial sets. If $f : X \rightarrow Y$ is a simplicial map such that each component $f_n : X_n \rightarrow Y_n$ is a bijection, then $f : X \rightarrow Y$ is a simplicial isomorphism.

2.5. Geometric Realisation of Simplicial Sets. One can construct a topological space from a simplicial set X by the so-called **geometric realisation**²:

$$|X| := \left(\bigsqcup_{n=0}^{\infty} X_n \times |\Delta^n| \right) / \sim,$$

where “ $\sim$ ” is the equivalence relation generated by the relations

$$\begin{cases} (x, D_i(p)) \sim (d_i(x), p) & \text{for } x \in X_{n+1}, p \in |\Delta^n|, \\ (x, S_i(p)) \sim (s_i(x), p) & \text{for } x \in X_{n-1}, p \in |\Delta^n|. \end{cases} \quad (2.5)$$

Here, we give X_n the discrete topology, then we take the topological sum (disjoint union) of products of topological spaces.

It is an exercise to check that the geometric realisation of the simplicial set Δ^n (Example 2.3.1) is exactly the geometric n -simplex $|\Delta^n|$ as we denoted on page 3.

Secondly, the geometric realisation is functorial: denote the equivalence class of $(x_n, t_n) \in X_n \times \Delta^n$ by $|x_n, t_n|$. Given a simplicial map $f : X \rightarrow Y$, the induced map $|f| : |X| \rightarrow |Y|$ is defined by

$$|f|(|x_n, t_n|) := |f_n(x_n), t_n|,$$

where f_n is the n -th component of the natural transformation $f : X \rightarrow Y$. We comment that $|f|$ is well-defined, which can be justified by using the simplicial identities and the fact that f is a natural transformation. Moreover, $|f|$ is continuous. This can shown by first noting that we have a map

$$\bigsqcup_{n \geq 0} X_n \times |\Delta^n| \rightarrow \bigsqcup_{n \geq 0} Y_n \times |\Delta^n|$$

defined by the disjoint union of the maps

$$(x_n, |\Delta^n|) \xrightarrow{(f_n, \text{id}_{|\Delta^n|})} (f_n(x_n), |\Delta^n|).$$

These maps are clearly continuous, and therefore give a continuous map between the disjoint unions. Then $|f|$ is just the induced map between quotient spaces, hence continuous as well. Therefore, we see that the geometric realisation is a functor

²[Fri08, page 24–25] gives a nice treatment of this definition.

$$|-| : \mathbf{sSet} \rightarrow \mathbf{Top}.$$

In fact, the following theorem of Milnor asserts that we actually get a nice topological space.

Theorem 2.5.1 ([Mil57, Theorem 1 on page 358])

For a simplicial set X , its geometric realisation $|X|$ is a CW-complex, having one n -cell corresponding to each non-degenerate n -simplex of X .

2.6. Examples of Geometric Realisations of Simplicial Sets.

2.6.1 Consider a topological space Y . There is a canonical map $|\mathcal{S}(Y)| \rightarrow Y$ defined by

$$|\sigma, t| := \sigma(t)$$

for $(\sigma, t) \in \mathcal{S}(Y)_n \times |\Delta^n|$. As noted in [Mil57, Theorem 4 on page 362], this map induces isomorphisms on homotopy groups and homology groups. If in addition Y is a CW-complex, then Whitehead Theorem and Theorem 2.5.1 imply that $|\mathcal{S}(Y)|$ and Y are homotopy equivalent.

2.6.2 Let $\mathcal{C}$ be a small category, and consider its nerve $\mathcal{N}(\mathcal{C})$ defined in Example 2.3.4. The **classifying space BC of $\mathcal{C}$** is the geometric realisation of its nerve:

$$BC := |\mathcal{N}(\mathcal{C})|.$$

Now let G be a group; and let $\underline{G}$ be the category with only one object e , and $\text{Mor}_{\underline{G}}(e, e) = G$, with composition performed by multiplying elements via the group operation. Then we have a natural homeomorphism

$$B\underline{G} \cong BG$$

between the classifying space of the category and of the group, for which the latter one is constructed by the bar construction. The homeomorphism is natural in the sense that there is a natural simplicial isomorphism between $\mathcal{N}(\underline{G})$ and the simplicial set obtained via the bar construction.

2.7. Milnor's Theorem on Geometric Realisation of Product of Simplicial Sets.

For simplicial sets X, Y , we define the **Milnor's map** $\eta : |X \times Y| \rightarrow |X| \times |Y|$ by

$$\eta(|x_n, y_n, t_n|) := (|x_n, t_n|, |y_n, t_n|) \quad (2.6)$$

We have the following theorem

Theorem 2.7.1 ([Mil57, Theorem 2 on page 359])

For two simplicial sets X, Y , if either

- (a) X, Y are both countable (i.e., X_n, Y_n are countable for all n), or
- (b) one of $|X|, |Y|$ is locally finite,

then η is a homeomorphism.

In general the two spaces $|X \times Y|$ and $|X| \times |Y|$ may not be homeomorphic, because $|X \times Y|$ is given the quotient topology inherited from $\bigsqcup_{n \geq 0} X_n \times \Delta^n$, while $|X| \times |Y|$ is given

the product topology. Readers can consult [Hat01, page 524–525] for such an example.

However, $|X|$ is always a **Kelley space (i.e., compactly generated Hausdorff space)**, see [GZ66, first paragraph on page 49] for a brief discussion. Some authors (e.g., [GJ10]) interpret geometric realisation as a functor taking values in the category of compactly generated Hausdorff space because of the following proposition:

Proposition 2.7.2 ([GJ10, Proposition 2.4 on page 9])

Let **CGHaus** be the category of Kelley spaces. Then the geometric realisation functor

$$|-| : \mathbf{sSet} \rightarrow \mathbf{CGHaus}$$

preserves finite limits.

2.8. $|-|$ and $\mathcal{S}(-)$ are Adjoint. In this section, we prove that $|-|$ and $\mathcal{S}(-)$ are adjoint functors:

Theorem 2.8.1

For any simplicial set X and topological space Y , there is a natural bijection between Hom-sets:

$$\mathrm{Hom}_{\mathbf{sSet}}(X, \mathcal{S}(Y)) \cong \mathrm{Hom}_{\mathbf{Top}}(|X|, Y).$$

Proof. Our proof is motivated by [Fri08, page 29].

For $f \in \mathrm{Hom}_{\mathbf{sSet}}(X, \mathcal{S}(Y))$, we define a continuous map $\Phi f \in \mathrm{Hom}_{\mathbf{Top}}(|X|, Y)$ by

$$\Phi f(|x_n, t_n|) := [f_n(x_n)](t_n) \in Y, \quad (2.7)$$

that is, the singular simplex $f_n(x_n) : |\Delta^n| \rightarrow Y$ evaluated at $t_n \in |\Delta^n|$.

For $g \in \mathrm{Hom}_{\mathbf{Top}}(|X|, Y)$, define a natural transformation $\Psi g = \{(\Psi g)_n\} \in \mathrm{Hom}_{\mathbf{sSet}}(X, \mathcal{S}(Y))$ by setting its components to be

$$(\Psi g)_n(x_n) := g(|x_n, -|) \in \mathrm{Hom}_{\mathbf{Top}}(|\Delta^n|, Y) \quad (2.8)$$

i.e., the value of the singular simplex $(\Psi g)_n(x_n)$ at $t_n \in |\Delta^n|$ is $g(|x_n, t_n|)$.

- (1). We first show that $\Phi : \mathrm{Hom}_{\mathbf{sSet}}(X, \mathcal{S}(Y)) \rightarrow \mathrm{Hom}_{\mathbf{Top}}(|X|, Y)$ and $\Psi : \mathrm{Hom}_{\mathbf{Top}}(|X|, Y) \rightarrow \mathrm{Hom}_{\mathbf{sSet}}(X, \mathcal{S}(Y))$ are inverses.

Indeed, for each $x_n \in X_n$, we have

$$\begin{aligned} [\Psi(\Phi f)]_n &:= \Phi f(|x_n, -|) \\ &:= f_n(x_n) \end{aligned}$$

Thus the n -th components of the natural transformations $\Psi(\Phi f)$, f agree, hence $\Psi(\Phi f) = f$.

On the other hand, for any $|x_n, t_n| \in |X|$ we have

$$\begin{aligned}\Phi\Psi g(|x_n, t_n|) &:= (\Psi g)_n(|x_n, t_n|) \\ &:= g(|x_n, t_n|)\end{aligned}$$

$$\therefore \Phi\Psi g = g.$$

- (2). Secondly, we show that the function $\Phi f : |X| \rightarrow Y$ is well-defined for any simplicial map $f : X \rightarrow \mathcal{S}(Y)$.

Write S be the simplex functor defined in Equation (2.2). If $|x_n, t_n| = |x_m, t_m|$, then there exists $\varphi \in \text{Hom}_\Delta([m], [n])$ such that

$$\begin{aligned}[X\varphi](x_n) &= x_m, \quad \text{and} \\ S\varphi(t_n) &= t_m.\end{aligned}$$

This follows from Lemma 2.1.1 and the relations defined in Equation (2.5).

Thus we have

$$\begin{aligned}\Phi f(|x_m, t_m|) &:= f_m(x_m)(t_m) \\ &= f_m([X\varphi](x_n))(t_m) \\ &= [\mathcal{S}(Y)\varphi](f_n(x_n)(t_m)) && \text{(the functor } \mathcal{S}(Y) \text{ applying to } \varphi.) \\ &= (f_n(x_n) \circ S\varphi)(t_m) && \text{(by definition of the functor } \mathcal{S}(Y).) \\ &= f_n(x_n)(S\varphi(t_m)) \\ &= f_n(x_n)(t_n) \\ &=: \Phi f(|x_n, t_n|).\end{aligned}$$

- (3). Thirdly, we show that $\Psi g = \{(\Psi g)_n\}$ is a simplicial map.

For $\varphi \in \text{Hom}_\Delta([m], [n])$, we wish to show that the diagram

$$\begin{array}{ccc} X_n & \xrightarrow{(\Psi g)_n} & \mathcal{S}(Y)_n \\ X\varphi \downarrow & & \downarrow \mathcal{S}(Y)\varphi \\ X_m & \xrightarrow{(\Psi g)_m} & \mathcal{S}(Y)_m \end{array}$$

commutes. Indeed, for any $x_n \in X_n$ and $t_m \in |\Delta^m|$ we compute

$$\begin{aligned}[\mathcal{S}(Y)\varphi \circ (\Psi g)_n(x_n)](t_m) &:= (\Psi g)_n(x_n)(S\varphi(t_m)) \\ &:= g(|x_n, S\varphi(t_m)|)\end{aligned}$$

$$\begin{aligned}
 &= g(|X\varphi(x_n), t_m|) \\
 &=: (\Psi g)_{n-1}(X\varphi(x_n))(t_m).
 \end{aligned}$$

$\therefore \Psi g$ is a simplicial map.

(4). At last but not least, we show the bijection is natural.

To show naturality in $Y \in \text{Obj}(\mathbf{Top})$, let $g \in \text{Hom}_{\mathbf{Top}}(Y, Y')$. Then we have an assignment $\mathcal{S}g : \text{Hom}_{\mathbf{sSet}}(X, \mathcal{S}(Y)) \rightarrow \text{Hom}_{\mathbf{sSet}}(X, \mathcal{S}(Y'))$ defined by

$$[(\mathcal{S}g)(h)]_n := (g \circ h_n : X_n \rightarrow \mathcal{S}(Y')_n)$$

for any $h \in \text{Hom}_{\mathbf{sSet}}(X, \mathcal{S}(Y))$. We wish to show the diagram

$$\begin{array}{ccc}
 \text{Hom}_{\mathbf{Top}}(|X|, Y) & \xrightarrow{g \circ -} & \text{Hom}_{\mathbf{Top}}(|X|, Y') \\
 \Psi \downarrow & & \downarrow \Psi \\
 \text{Hom}_{\mathbf{sSet}}(X, \mathcal{S}(Y)) & \xrightarrow{\mathcal{S}g} & \text{Hom}_{\mathbf{sSet}}(X, \mathcal{S}(Y'))
 \end{array}$$

commutes. For any $f \in \text{Hom}_{\mathbf{Top}}(|X|, Y)$ we compute

$$\begin{aligned}
 [\Psi(g \circ f)]_n(x_n) &:= (g \circ f)(|x_n, -|) \\
 &= g(f(|x_n, -|)) \\
 &=: [g \circ (\Psi f)]_n(x_n) \\
 &=: [\mathcal{S}g(\Psi f)]_n(x_n).
 \end{aligned}$$

For the naturality in $X \in \text{Obj}(\mathbf{sSet})$, let $r \in \text{Hom}_{\mathbf{sSet}}(X', X)$. We wish to show the diagram

$$\begin{array}{ccc}
 \text{Hom}_{\mathbf{sSet}}(X, \mathcal{S}(Y)) & \xrightarrow{- \circ r} & \text{Hom}_{\mathbf{sSet}}(X', \mathcal{S}(Y)) \\
 \Phi \downarrow & & \downarrow \Phi \\
 \text{Hom}_{\mathbf{Top}}(|X|, Y) & \xrightarrow{- \circ |r|} & \text{Hom}_{\mathbf{Top}}(|X'|, Y)
 \end{array}$$

commutes. For any $p \in \text{Hom}_{\mathbf{sSet}}(X, \mathcal{S}(Y))$ we compute

$$\begin{aligned}
 [\Phi(p \circ r)](|x'_n, t_n|) &:= [(p \circ r)_n(x'_n)](t_n) \\
 &= [(p_n \circ r_n)(x'_n)](t_n) \\
 &= [p_n(r_n(x'_n))](t_n) \\
 &=: \Phi p(|r_n(x'_n), t_n|) \\
 &=: (\Phi p \circ |r|)(|x'_n, t_n|).
 \end{aligned}$$

This concludes that $\text{Hom}_{\mathbf{sSet}}(X, \mathcal{S}(Y))$ and $\text{Hom}_{\mathbf{Top}}(|X|, Y)$ are naturally in bijection. $\square$

2.9. The Goerss-Jardine Formalism. In this section we introduce another approach to formulate geometric realisation of a simplicial set, after Goerss-Jardine [GJ10]. An advantage of this approach is that the proof for Theorem 2.8.1 will only be 4 lines. The disadvantage, however, is that we need to go deeper into category theory. In particular, Yoneda's Lemma is required.

The materials covered in this section are from [GJ10, page 3–7].

We first introduce the Yoneda's Lemma for simplicial sets.

Theorem 2.9.1 (Yoneda's Lemma for Simplicial Sets)

Let X be a simplicial set. There is natural bijection

$$X_n \cong \text{Hom}_{\mathbf{sSet}}(\Delta^n, X).$$

This is an instance of the usual Yoneda's Lemma ([ML98, page Lemma on 61]).

Now, for a simplicial set X , we can form the **simplex category** $\Delta \downarrow X$, whose objects are simplicial maps

$$\sigma : \Delta^n \rightarrow X,$$

and morphisms are commutative diagrams

$$\begin{array}{ccc} \Delta^n & \xrightarrow{\sigma} & X \\ \theta \downarrow & \nearrow \tau & \\ \Delta^m & & \end{array}.$$

Composition of morphisms are performed in the obvious way. Next, there exists a functor $\Delta \downarrow X \rightarrow \mathbf{sSet}$ defined by

$$(\sigma : \Delta^n \rightarrow X) \mapsto \Delta^n,$$

and

$$\left(\begin{array}{ccc} \Delta^n & \xrightarrow{\sigma} & X \\ \theta \downarrow & \nearrow \tau & \\ \Delta^m & & \end{array} \right) \mapsto (\theta : \Delta^n \rightarrow \Delta^m).$$

Then we have the following result

Proposition 2.9.2 ([GJ10, Lemma 2.1 on page 7])

A simplicial set X is naturally isomorphic to the colimit of the functor:

$$X \cong \operatorname{colim}_{\Delta \downarrow X} \Delta^n.$$

Now we have a sequence of functors

$$\Delta \downarrow X \rightarrow \mathbf{sSet} \xrightarrow{\text{geometric realisation}} \mathbf{Top}.$$

The **Goerss-Jardine realisation** of a simplicial set X is defined to be the colimit of the composition functor:

$$|X|_{\text{GJ}} := \operatorname{colim}_{\Delta \downarrow X} |\Delta^n|,$$

and has the adjoint property with singular set:

Theorem 2.9.3 ([GJ10, Proposition 2.2 on page 7])

For any simplicial set X and topological set Y , there is a natural bijection between Hom-sets:

$$\operatorname{Hom}_{\mathbf{sSet}}(X, \mathcal{S}(Y)) \cong \operatorname{Hom}_{\mathbf{Top}}(|X|_{\text{GJ}}, Y).$$

Proof.

$$\begin{aligned} \operatorname{Hom}_{\mathbf{Top}}(|X|_{\text{GJ}}, Y) &:= \operatorname{Hom}_{\mathbf{Top}}\left(\operatorname{colim}_{\Delta \downarrow X} |\Delta^n|, Y\right) \\ &\cong \lim_{\Delta \downarrow X} \operatorname{Hom}_{\mathbf{Top}}(|\Delta^n|, Y) \\ &:= \lim_{\Delta \downarrow X} \mathcal{S}(Y)_n \\ &\cong \lim_{\Delta \downarrow X} \operatorname{Hom}_{\mathbf{sSet}}(\Delta^n, \mathcal{S}(Y)) \quad (\text{Yoneda's Lemma, Theorem 2.9.1}) \\ &\cong \operatorname{Hom}_{\mathbf{sSet}}\left(\operatorname{colim}_{\Delta \downarrow X} \Delta^n, Y\right) \\ &\cong \operatorname{Hom}_{\mathbf{sSet}}(X, \mathcal{S}(Y)) \quad (\text{Proposition 2.9.2}). \end{aligned}$$

Moreover, all isomorphisms presented here are natural either by earlier results or elementary facts from category theory. Thus the claim holds as desired. $\square$

The Goerss-Jardine realisation turns out to be homeomorphic to the geometric realisation.

Theorem 2.9.4

Let X be a simplicial set. Then

$$|X| \cong |X|_{\text{GJ}}.$$

Proof. Consider the following colimit diagram in the category $\Delta \downarrow X$:

$$\begin{array}{ccccccc}
\cdots & \longrightarrow & \Delta^m & \xrightarrow{\theta_{mn}} & \Delta^n & \xrightarrow{\theta_{n\ell}} & \Delta^\ell \longrightarrow \cdots \\
& & \searrow \sigma_m & & \downarrow \sigma_n & & \swarrow \sigma_\ell \\
& & & & X & &
\end{array}$$

By the definition of $\Delta \downarrow X$, we know that each triangle commutes. Next, since geometric realisation is functorial, we have the following commutative diagram:

$$\begin{array}{ccccccc}
\cdots & \longrightarrow & |\Delta^m| & \xrightarrow{|\theta_{mn}|} & |\Delta^n| & \xrightarrow{|\theta_{n\ell}|} & |\Delta^\ell| \longrightarrow \cdots \\
& & \searrow |\sigma_m| & & \downarrow |\sigma_n| & & \swarrow |\sigma_\ell| \\
& & & & |X| & &
\end{array}$$

It remains to show that if Y is a topological space such that the following diagram

$$\begin{array}{ccccccc}
\cdots & \longrightarrow & |\Delta^m| & \xrightarrow{|\theta_{mn}|} & |\Delta^n| & \xrightarrow{|\theta_{n\ell}|} & |\Delta^\ell| \longrightarrow \cdots \\
& & \searrow \tau_m & & \downarrow \tau_n & & \swarrow \tau_\ell \\
& & & & Y & &
\end{array}$$

commutes, then there exists a unique map $f : |X| \rightarrow Y$ so that the diagram

$$\begin{array}{ccccccc}
\cdots & \longrightarrow & |\Delta^m| & \xrightarrow{|\theta_{mn}|} & |\Delta^n| & \xrightarrow{|\theta_{n\ell}|} & |\Delta^\ell| \longrightarrow \cdots \\
& & \searrow |\sigma_m| & & \downarrow |\sigma_n| & & \swarrow |\sigma_\ell| \\
& & & & |X| & & \\
& \searrow \tau_m & & & \downarrow f & & \swarrow \tau_\ell \\
& & & & Y & &
\end{array} \tag{2.9}$$

commutes. Notice that it suffices to find a map f_2 from the disjoint union $\bigsqcup_{n \geq 0} X_n \times |\Delta^n|$ to Y ,

and show that it respects the equivalence relations in $|X|$. Then by the universal property of quotient, f will be the induced map of f_2 . But then, by the universal property of coproduct (the disjoint union), one only needs to define maps $f_3 : X_n \times |\Delta^n| \rightarrow Y$ for each n in order to get f_2 ; and there is a preferred choice for f_3 , namely f_3 is the composition

$$X_n \times |\Delta^n| \xrightarrow{\text{projection}} |\Delta^n| \xrightarrow{\tau_n} Y.$$

Therefore, we have a induced map $f : |X| \rightarrow Y$. We leave readers to verify that f is the unique that makes diagram (2.9) commutes.

□

We want to comment that Theorem 2.9.4 can be thought as an immediate consequence of the **Special Adjoint Functor Theorem** ([ML98, Theorem 2 on page 129]³): since the geometric realisation functor $|\cdot|$ has a right adjoint (Theorem 2.8.1), it commutes with colimits. Hence

$$\begin{aligned} |X|_{\text{GJ}} &:= \operatorname{colim}_{\Delta \downarrow X} |\Delta^n| \\ &\cong \left| \operatorname{colim}_{\Delta \downarrow X} \Delta^n \right| \\ &\cong |X|. \end{aligned}$$

2.10. Colimit of Simplicial Sets. Let $\mathbf{C}$ be a small category, and $F : \mathbf{C} \rightarrow \mathbf{sSet}$ be a functor. For each $c \in \operatorname{Obj}(\mathbf{C})$, write $X_c = F(c)$. We then can form the **colimit of simplicial sets** $\operatorname{colim}_{c \in \operatorname{Obj}(\mathbf{C})} X_c$ by taking

$$\left(\operatorname{colim}_{c \in \operatorname{Obj}(\mathbf{C})} X_c \right)_n := \operatorname{colim}_{c \in \operatorname{Obj}(\mathbf{C})} (X_c)_n.$$

The i -th face operator d_i is defined as follow: any n -simplex x of the colimit is represented by some n -simplex x_α of X_α . We apply $d_i^{X_\alpha}$ to x_α to get an $(n-1)$ -simplex x'_α of X_α . Then map x'_α to $\operatorname{colim}_{c \in \operatorname{Obj}(\mathbf{C})} (X_c)_{n-1}$, and the resulting $(n-1)$ -simplex of the colimit is $d_i(x)$.

More precisely, we have a commutative ladder,

$$\begin{array}{ccccccc} x_\alpha & \xrightarrow{\quad\quad\quad} & & & & & x \\ \cap & & & & & & \cap \\ \cdots & \longrightarrow & (X_\alpha)_n & \longrightarrow & (X_\beta)_n & \longrightarrow & (X_\gamma)_n \longrightarrow \cdots \longrightarrow \operatorname{colim}_{c \in \operatorname{Obj}(\mathbf{C})} (X_c)_n \\ & & \downarrow d_i^{X_\alpha} & & \downarrow d_i^{X_\beta} & & \downarrow d_i^{X_\gamma} & & \downarrow d_i \\ \cdots & \longrightarrow & (X_\alpha)_{n-1} & \longrightarrow & (X_\beta)_{n-1} & \longrightarrow & (X_\gamma)_{n-1} \longrightarrow \cdots \longrightarrow \operatorname{colim}_{c \in \operatorname{Obj}(\mathbf{C})} (X_c)_{n-1} \\ \cup & & & & & & \cup \\ x'_\alpha = d_i^{X_\alpha}(x_\alpha) & \xrightarrow{\quad\quad\quad} & & & & & d_i x \end{array}$$

in which the horizontal arrows are morphisms that used to define the n -simplex of the colimit, vertical arrows are face maps for each simplicial sets in the directed system. The vertical arrow on far right is defined to be the induced map of colimits.

The degeneracy maps are defined in the same fashion, and we let the readers to verify that they are well-defined, and satisfy the simplicial identities.

³We use the dual version of this theorem. i.e., the one for colimit.

3. BISIMPLICIAL SETS

3.1. Definition of Bisimplicial Set. A **bisimplicial set** is a functor $X : \Delta^{op} \times \Delta^{op} \rightarrow \mathbf{Set}$, together with vertical face and degeneracy maps

$$\begin{aligned} d_i^v : X_{m,n} &\rightarrow X_{m,n-1}, \\ s_i^v : X_{m,n} &\rightarrow X_{m,n+1}, \end{aligned}$$

and horizontal face and degeneracy maps

$$\begin{aligned} d_i^h : X_{m,n} &\rightarrow X_{m-1,n}, \\ s_i^h : X_{m,n} &\rightarrow X_{m+1,n}, \end{aligned}$$

such that vertical maps and horizontal maps satisfy the simplicial identities on their own, and vertical maps commute with horizontal maps.

We call $X_{m,n} := X([m], [n])$ the **(m, n)-simplex** of X .

Just as in the case of simplicial set, a **bisimplicial map** between two bisimplicial sets is just a natural transformation $f : X \rightarrow Y$, and we denote by **bsSet** the **category of bisimplicial sets**.

3.2. Examples of Bisimplicial Sets.

3.2.1 The **bisimplex** $\Delta^{m,n}$ is defined to be the functor

$$([p], [q]) \mapsto \Delta_p^m \times \Delta_q^n$$

with

$$\left\{ \begin{array}{l} d_j^h : \Delta_p^m \times \Delta_q^n \xrightarrow{d_j^{\Delta^m}} \Delta_{p-1}^m \times \Delta_q^n, \\ s_j^h : \Delta_p^m \times \Delta_q^n \xrightarrow{s_j^{\Delta^m}} \Delta_{p+1}^m \times \Delta_q^n, \end{array} \right. \quad \left\{ \begin{array}{l} d_j^v : \Delta_p^m \times \Delta_q^n \xrightarrow{d_j^{\Delta^n}} \Delta_p^m \times \Delta_{q-1}^n, \\ s_j^v : \Delta_p^m \times \Delta_q^n \xrightarrow{s_j^{\Delta^n}} \Delta_p^m \times \Delta_{q+1}^n. \end{array} \right. \quad (3.1)$$

as defined in Example 2.3.1.

3.2.2 (Bisingular Set) It should not be surprising that there exists a bisimplicial version of the singular set. The **bisingular set** $b\mathcal{S}(Y)$ of a topological space Y has (p, q) -simplices:

$$b\mathcal{S}(Y)_{p,q} := \{ \sigma : |\Delta^p| \times |\Delta^q| \rightarrow Y \mid \sigma \text{ is continuous.} \},$$

with

$$\begin{cases} d_j^h : b\mathcal{S}(Y)_{p,q} \xrightarrow{d_j^{\mathcal{S}(Y)}} b\mathcal{S}(Y)_{p-1,q}, \\ s_j^h : b\mathcal{S}(Y)_{p,q} \xrightarrow{s_j^{\mathcal{S}(Y)}} b\mathcal{S}(Y)_{p+1,q}, \end{cases} \quad (3.2)$$

$$\begin{cases} d_j^v : b\mathcal{S}(Y)_{p,q} \xrightarrow{d_j^{\mathcal{S}(Y)}} b\mathcal{S}(Y)_{p,q-1}, \\ s_j^v : b\mathcal{S}(Y)_{p,q} \xrightarrow{s_j^{\mathcal{S}(Y)}} b\mathcal{S}(Y)_{p,q+1}. \end{cases}$$

as defined in Example 2.3.2.

3.2.3 (External Product) Given two simplicial sets X, Y we can form a bisimplicial set by taking their **external product** :

$$(X \tilde{\times} Y)_{p,q} := X_p \times Y_q,$$

with

$$\begin{cases} d_j^h : X_p \times Y_q \xrightarrow{d_j^X} X_{p-1} \times Y_q, \\ s_j^h : X_p \times Y_q \xrightarrow{s_j^X} X_{p+1} \times Y_q, \end{cases} \quad \begin{cases} d_j^v : X_p \times Y_q \xrightarrow{d_j^Y} X_p \times Y_{q-1}, \\ s_j^v : X_p \times Y_q \xrightarrow{s_j^Y} X_p \times Y_{q+1}, \end{cases} \quad (3.3)$$

The bisimplex $\Delta^{m,n}$ can then be re-written as the external product

$$\Delta^{m,n} = \Delta^m \tilde{\times} \Delta^n.$$

In particular, (the general) Yoneda's Lemma implies that $\Delta^{m,n}$ is representable:

$$\begin{aligned} \Delta^{m,n} &\cong \text{Hom}_{\Delta \times \Delta}(-, [m] \times [n]) \\ &\cong \text{Hom}_{\Delta}(-, [m]) \times \text{Hom}_{\Delta}(-, [n]) \end{aligned}$$

and hence

$$X_{m,n} \cong \text{Hom}_{\mathbf{bsSet}}(\Delta^{m,n}, X)$$

for any bisimplicial set X .

3.2.4 The colimit of bisimplicial sets is defined in a similar manner as in Section 2.10.

3.3. Bisimplex Category and Diagonal Simplex Category. Like in 2.9, we can form the so-called **bisimplex category** $b\Delta \downarrow X$ for a bisimplicial set X . Its objects are bisimplicial maps

$$\Delta^{m,n} \xrightarrow{\sigma} X,$$

and morphisms are commutative diagrams

$$\begin{array}{ccc}
\Delta^{m,n} & \xrightarrow{\sigma} & X \\
\theta \downarrow & \nearrow \tau & \\
\Delta^{p,q} & &
\end{array}$$

We then have an analogue of Proposition 2.9.2 for bisimplicial sets:

Proposition 3.3.1

For a bisimplicial set X , there is a natural isomorphism

$$X \cong \operatorname{colim}_{b\Delta \downarrow X} \Delta^{m,n}.$$

Bisimplex category will be useful when studying the the topological aspects of a bisimplicial set.

4. TWO GEOMETRIC REALISATIONS OF BISIMPLICIAL SETS

Of course, there exists a (functorial) way to produce a topological space from a bisimplicial set. In fact, there are at least four of them. We present two here.

4.1. Full Realisation. The **full realisation** of a bisimplicial set X is defined to be

$$|X|_{\text{full}} := \left(\bigsqcup_{m,n \geq 0} X_{m,n} \times |\Delta^m| \times |\Delta^n| \right) / \sim, \quad (4.1)$$

where $X_{m,n}$ is equipped with the discrete topology as usual, and “ $\sim$ ” is the equivalence relation generated by

$$\left\{ \begin{array}{ll}
(x, D_i(t), s) \sim (d_i^h(x), t, s) & \text{for } x \in X_{m+1,n}, t \in |\Delta^m|, s \in |\Delta^n|, \\
(x, S_i(t), s) \sim (s_i^h(x), t, s) & \text{for } x \in X_{m-1,n}, t \in |\Delta^m|, s \in |\Delta^n|, \\
(x, t, D_i(s)) \sim (d_i^v(x), t, s) & \text{for } x \in X_{m,n+1}, t \in |\Delta^m|, s \in |\Delta^n|, \\
(x, t, S_i(s)) \sim (s_i^v(x), t, s) & \text{for } x \in X_{m,n-1}, t \in |\Delta^m|, s \in |\Delta^n|.
\end{array} \right. \quad (4.2)$$

Proposition 4.1.1

$|\Delta^{p,q}|_{\text{full}}$ is homeomorphic to $|\Delta^p| \times |\Delta^q|$.

Proof.

$$\begin{aligned}
|\Delta^{p,q}|_{\text{full}} &:= \left(\bigsqcup_{m,n \geq 0} \Delta_{m,n}^{p,q} \times |\Delta^m| \times |\Delta^n| \right) / \sim \\
&:= \left(\bigsqcup_{m,n \geq 0} \Delta_m^p \times \Delta_n^q \times |\Delta^m| \times |\Delta^n| \right) / \sim
\end{aligned}$$

$$\begin{aligned}
 &\cong \left\{ \bigsqcup_{n \geq 0} \left[\left(\bigsqcup_{m \geq 0} \Delta_m^p \times |\Delta^m| \right) / \left(\text{horizontal relations} \right) \right] \times \Delta_n^q \times |\Delta^n| \right\} / \left(\text{vertical relations} \right) \\
 &= \left\{ \bigsqcup_{n \geq 0} |\Delta^p| \times \Delta_n^q \times |\Delta^n| \right\} / \left(\text{vertical relations} \right) \\
 &\cong |\Delta^p| \times \left[\left(\bigsqcup_{n \geq 0} \Delta_n^q \times |\Delta^n| \right) / \left(\text{vertical relations} \right) \right] \\
 &= |\Delta^p| \times |\Delta^q|
 \end{aligned}$$

We comment that the reason why we can separate the horizontal and vertical relations in such way is because of the universal property of quotient space. $\square$

Notice that we only need to modify the proof a little bite to obtain a more general statement when a bisimplicial set is an external product of two simplicial sets.

Addendum 4.1.2

Let X be a bisimplicial set. If there exists simplicial sets Y, Z such that $X = Y \tilde{\times} Z$, then $|X|_{\text{full}}$ is homeomorphic to $|Y| \times |Z|$.

4.2. Diagonal Realisation. We saw earlier in Example 3.2.3 that one can form bisimplicial sets from simplicial sets. There is a construction in the reverse direction.

Let X be a bisimplicial set, its **diagonal** $\text{diag}(X)$ is the simplicial set given by

$$\text{diag}(X)_n := X_{n,n},$$

and

$$d_j^{\text{diag}(X)} := d_j^h d_j^v (= d_j^v d_j^h), \quad s_j^{\text{diag}(X)} = s_j^h s_j^v (= s_j^v s_j^h).$$

In other words, $\text{diag}(X) : \Delta^{op} \rightarrow \mathbf{Set}$ is the functor obtained by composing the following functors

$$\Delta^{op} \xrightarrow[\substack{\text{diagonal} \\ \text{functor}}]{[n] \mapsto ([n], [n])} \Delta^{op} \times \Delta^{op} \xrightarrow{X} \mathbf{Set}.$$

The **diagonal realisation** of a bisimplicial set X is given by

$$|X|_{\text{diag}} := |\text{diag}(X)|, \tag{4.3}$$

that is, the geometric realisation of the simplicial set $\text{diag}(X)$.

4.3. Examples. Consider $\Delta^{p,q}$. We see that

$$\begin{aligned}
 \text{diag}(\Delta^{p,q})_n &:= \Delta_{n,n}^{p,q} \\
 &= \Delta_n^p \times \Delta_n^q \\
 &= (\Delta^p \times \Delta^q)_n
 \end{aligned}$$

Therefore, the diagonal of the bisimplicial set $\Delta^{p,q}$ is the product of the simplicial sets $\Delta^p \times \Delta^q$.

Similarly, if a bisimplicial set X can be written as an external product $X = Y \tilde{\times} Z$ of simplicial sets Y, Z , then $\text{diag}(X) = Y \times Z$. As a result, its diagonal realisation is

$$\begin{aligned} |X|_{\text{diag}} &= |Y \tilde{\times} Z|_{\text{diag}} \\ &= |\text{diag}(Y \tilde{\times} Z)| \\ &= |Y \times Z| \\ &= |Y| \times |Z| \end{aligned} \quad (\text{by Milnor's Theorem 2.7.1})$$

The diagonal realisation of a bisimplicial set can be computed from the elementary pieces:

Proposition 4.3.1

Let X be a bisimplicial set. Then

- (i) $\text{diag}(X) \cong \text{colim}_{b\Delta \downarrow X} [\text{diag}(\Delta^{m,n})]$,
- (ii) $|X|_{\text{diag}} \cong \text{colim}_{b\Delta \downarrow X} (|\Delta^m| \times |\Delta^n|)$.

Proof. (i)

$$\begin{aligned} \left(\text{colim}_{b\Delta \downarrow X} \text{diag}(\Delta^{m,n}) \right)_p &:= \text{colim}_{b\Delta \downarrow X} [\text{diag}(\Delta^{m,n})]_p && (\text{See Example 3.2.4}) \\ &:= \text{colim}_{b\Delta \downarrow X} \Delta_p^m \times \Delta_p^n \\ &= \text{colim}_{b\Delta \downarrow X} (\Delta^m \tilde{\times} \Delta^n)_{pp} \\ &=: \left(\text{colim}_{b\Delta \downarrow X} \Delta^{m,n} \right)_{pp} \\ &\cong X_{pp} && (\text{Proposition 3.3.1}) \\ &=: \text{diag}(X)_p. \end{aligned}$$

$\therefore$ By Lemma 2.4.1, (i) holds.

(ii)

$$\begin{aligned} |X|_{\text{diag}} &:= |\text{diag}(X)| \\ &\cong \left| \text{colim}_{b\Delta \downarrow X} \text{diag}(\Delta^{m,n}) \right| \\ &= \text{colim}_{b\Delta \downarrow X} |\text{diag}(\Delta^{m,n})| && (\text{Special Adjoint Functor Theorem}) \\ &= \text{colim}_{b\Delta \downarrow X} (|\Delta^m| \times |\Delta^n|) && (\text{Example 4.3}) \end{aligned}$$

□

Together with the Special Adjoint Functor Theorem, Proposition 4.3.1 implies that the diagonal functor

$$\text{diag}(-) : \mathbf{bsSet} \rightarrow \mathbf{sSet}$$

admits a left adjoint. It will be interesting to construct the left adjoint explicitly.

5. ADJUNCTIONS FOR BISIMPLICIAL SETS

Theorem 5.0.1

For any bisimplicial set X and topological space Y , there are natural bijections between Hom-sets:

- (i). $\text{Hom}_{\mathbf{bsSet}}(X, b\mathcal{S}(Y)) \cong \text{Hom}_{\mathbf{Top}}(|X|_{\text{full}}, Y),$
- (ii). $\text{Hom}_{\mathbf{bsSet}}(X, b\mathcal{S}(Y)) \cong \text{Hom}_{\mathbf{Top}}(|X|_{\text{diag}}, Y).$

Proof. (i). We will mimic the proof of Theorem 2.8.1.

For $f \in \text{Hom}_{\mathbf{bsSet}}(X, b\mathcal{S}(Y))$, we define a continuous map $\Phi f \in \text{Hom}_{\mathbf{Top}}(|X|_{\text{full}}, Y)$ by

$$\Phi f(|x_{mn}, t_m, s_n|) := [f_{mn}(x_{mn})](t_m, s_n) \in Y. \quad (5.1)$$

For $g \in \text{Hom}_{\mathbf{Top}}(|X|_{\text{full}}, Y)$, define a natural transformation $\Psi g = \{(\Psi g)_{mn}\} \in \text{Hom}_{\mathbf{bsSet}}(X, b\mathcal{S}(Y))$ by setting its components to be

$$(\Psi g)_{mn}(x_{mn}) := g(|x_{mn}, -, -|) \in \text{Hom}_{\mathbf{Top}}(|\Delta^m| \times |\Delta^n|, Y) \quad (5.2)$$

We let readers to verify (just as in Theorem 2.8.1) that Φ and Ψ are well-defined, inverses, and natural.

(ii). This is much easier. We have

$$\begin{aligned} \text{Hom}_{\mathbf{Top}}(|X|_{\text{diag}}, Y) &\cong \text{Hom}_{\mathbf{Top}}\left(\text{colim}_{b\Delta \downarrow X}(|\Delta^m| \times |\Delta^n|), Y\right) && \text{(Proposition 4.3.1)} \\ &\cong \text{Hom}_{\mathbf{Top}}\left(\text{colim}_{b\Delta \downarrow X}|\Delta^{m,n}|_{\text{full}}, Y\right) && \text{(Proposition 4.1.1)} \\ &\cong \lim_{b\Delta \downarrow X} \text{Hom}_{\mathbf{Top}}(|\Delta^{m,n}|_{\text{full}}, Y) \\ &\cong \lim_{b\Delta \downarrow X} \text{Hom}_{\mathbf{bsSet}}(\Delta^{m,n}, b\mathcal{S}(Y)) && \text{(by (i))} \\ &\cong \text{Hom}_{\mathbf{bsSet}}\left(\text{colim}_{b\Delta \downarrow X} \Delta^{m,n}, b\mathcal{S}(Y)\right) \\ &\cong \text{Hom}_{\mathbf{bsSet}}(X, b\mathcal{S}(Y)) \end{aligned}$$

All isomorphisms are natural, so we are done. □

We close our notes by showing that the full realisation and diagonal realisation of a bisimplicial set are the same:

Corollary 5.0.2

For a bisimplicial set X , we have

$$|X|_{\text{full}} \cong |X|_{\text{diag}}.$$

Proof. We saw that two realisations of a bisimplicial set X share the same right adjoint. Hence by uniqueness of adjoint functor [ML98, Corollary 1 on page 85], we know $|X|_{\text{full}} \cong |X|_{\text{diag}}$. $\square$

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